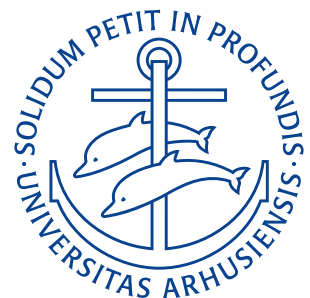


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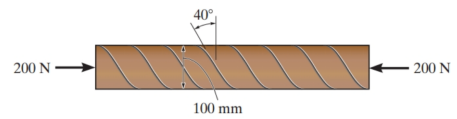
Take-Home Assignment Lec. 24–27

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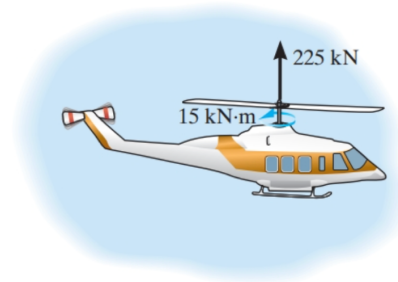
Problem 14.22

A paper tube is formed by rolling a cardboard strip in a spiral and then gluing the edges together as shown. Determine the shear stress acting along the seam, which is at 40° from the vertical, when the tube is subjected to an axial compressive force of 200 N. The paper is 2 mm thick and the tube has an outer diameter of 100 mm.



Problem 14.45

The rotor shaft of the helicopter is subjected to the tensile force and torque shown when the rotor blades provide the lifting force to suspend the helicopter at midair. If the shaft has a diameter of 150 mm, determine the principal stress and maximum in-plane shear stress at a point located on the surface of the shaft.



Problem 14.51

The stress at a point is shown on the element. Determine the principal stress and the absolute maximum shear stress.

